Manufacturing Process Health and Operational Efficiency Analysis in 6G-Enabled Smart Factories

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***Abstract*—The rapid advancement of Industry 4.0 technologies has significantly increased the importance of smart manufacturing systems and Industrial Internet of Things (IIoT) environments. Modern smart factories generate large volumes of operational and sensor data through interconnected industrial machines and high-speed communication systems. However, manufacturing teams often struggle to convert this data into meaningful operational insights for machine health monitoring, production optimization, and quality management. This paper presents a Manufacturing Process Health and Operational Efficiency Analysis system for 6G-enabled smart factories. The objective of the paper is to analyze machine-level sensor data, production performance metrics, quality indicators, predictive maintenance scores, and network communication parameters in order to identify operational inefficiencies, unhealthy machines, and quality bottlenecks. The dataset includes industrial parameters such as temperature, vibration, power consumption, production speed, defect rate, error rate, predictive maintenance score, network latency, and packet loss. Multiple analytical techniques and industrial Key Performance Indicators (KPIs) were developed to evaluate manufacturing performance and machine operating conditions. An interactive Streamlit dashboard was developed to visualize machine health, production quality, efficiency distribution, and 6G network performance in real time. The system supports centralized operational visibility and enables data-driven decision-making in modern smart manufacturing environments.**

***Index Terms*—Smart Manufacturing, Industry 4.0, 6G Networks, Industrial IoT, Machine Health Analysis, Streamlit Dashboard, Operational Efficiency, Manufacturing Analytics**

I. Introduction

The manufacturing industry is rapidly evolving due to advancements in Industrial Internet of Things (IIoT), artificial intelligence, automation, robotics, and high-speed communication technologies. Modern smart factories rely on interconnected industrial machines and intelligent communication systems to improve operational efficiency, production quality, and manufacturing reliability. In Industry 4.0 environments, industrial machines continuously generate sensor data related to temperature, vibration, energy consumption, production output, predictive maintenance conditions, and operational quality metrics. This large volume of industrial data creates opportunities for advanced manufacturing analytics and intelligent operational diagnostics.

Despite the availability of machine-generated data, many manufacturing organizations face challenges in monitoring machine health, correlating sensor behavior with production efficiency, identifying quality bottlenecks, and detecting operational degradation in real time. Reactive issue handling often results in machine downtime, reduced production output, increased maintenance costs, and lower manufacturing quality. The introduction of 6G-enabled communication systems further increases the complexity of industrial operations. Parameters such as network latency and packet loss directly influence industrial synchronization, machine coordination, and real-time data communication. This paper addresses these challenges by developing a manufacturing analytics dashboard capable of analyzing machine health, operational efficiency, quality behavior, and network communication performance. The developed system provides centralized visibility into smart factory operations and supports data-driven operational decision-making.

II. Literature Review

The advancement of Industry 4.0 has transformed traditional manufacturing systems into intelligent and highly connected smart factory environments. These environments integrate Industrial Internet of Things (IIoT) devices, cyber-physical systems, artificial intelligence, cloud computing, and advanced communication technologies to improve operational efficiency and manufacturing performance.

A first group of studies focuses on the foundational technologies of Industry 4.0 and Industrial IoT. Xu et al. [1] presented one of the earliest comprehensive surveys on IoT applications in industrial environments, highlighting the role of interconnected sensors and devices in enabling intelligent manufacturing. Similarly, Lu [2] provided a broad overview of Industry 4.0 technologies and identified smart manufacturing, big data analytics, cloud computing, and intelligent automation as key drivers of future industrial systems. Zhou et al. [10] further discussed the opportunities and challenges associated with Industry 4.0 adoption and emphasized the importance of digital transformation in manufacturing industries. While these studies establish the technological foundations of smart manufacturing, they primarily provide conceptual and architectural perspectives rather than operational diagnostic frameworks for monitoring manufacturing performance.

A second group of research focuses on cyber-physical systems and intelligent manufacturing architectures. Lee et al. [4] proposed a cyber-physical systems architecture that integrates physical manufacturing processes with digital monitoring and control systems. Zhong et al. [7] reviewed intelligent manufacturing technologies and emphasized the importance of integrating data analytics, automation, and smart decision-making mechanisms into modern production environments. These studies demonstrate how intelligent architectures can enhance manufacturing operations; however, they do not provide detailed mechanisms for simultaneously evaluating machine health, production efficiency, quality behavior, and communication performance within a unified analytics platform.

Another important research direction involves data-driven manufacturing and industrial analytics. Tao et al. [8] introduced the concept of data-driven smart manufacturing and demonstrated how large-scale industrial data can be transformed into actionable operational intelligence. Wan et al. [3] investigated large-scale IoT data processing and resource management techniques, highlighting the importance of efficient analytics for extracting value from industrial sensor data. Although these works emphasize the importance of industrial data analytics, they primarily focus on data processing and information extraction rather than developing operational dashboards capable of providing real-time manufacturing diagnostics. Machine health monitoring and predictive maintenance have also received significant research attention. Carvalho et al. [5] reviewed machine learning applications for predictive maintenance and demonstrated how sensor-driven analytics can improve machine reliability and reduce unexpected failures. Their work highlights the growing importance of monitoring operational parameters such as vibration, temperature, and power consumption for maintenance planning. However, predictive maintenance studies generally focus on failure prediction and maintenance scheduling, while providing limited insight into production quality, efficiency status, and communication performance metrics. Industrial communication infrastructure represents another critical component of smart manufacturing environments. Wollschlaeger et al. [6] examined industrial communication systems and emphasized the importance of reliable network infrastructures for Industry 4.0 applications. More recently, Mahmood et al. [9] presented a vision for future 6G communication systems and discussed how ultra-reliable low-latency communication can support advanced industrial automation and smart factory applications. While these studies address communication technologies and network architectures, they do not investigate how communication performance indicators such as latency and packet loss interact with manufacturing efficiency and machine behavior.

Based on the literature, several research gaps can be identified. First, most existing studies focus on individual aspects of smart manufacturing, such as IoT architectures, predictive maintenance, industrial communication, or data analytics, without providing a unified framework for operational diagnostics. Second, limited research has explored the combined analysis of machine health indicators, production performance metrics, quality inspection parameters, and network communication characteristics within a single monitoring platform. Third, despite the growing interest in 6G-enabled manufacturing systems, few studies incorporate communication metrics such as latency and packet loss into manufacturing performance analysis. Finally, many existing works emphasize predictive or architectural frameworks rather than practical diagnostic dashboards that provide centralized visibility into factory operations.

To address these gaps, this paper develops a Manufacturing Process Health and Operational Efficiency Analysis framework for 6G-enabled smart factories. Unlike previous studies, the proposed system integrates machine health monitoring, production efficiency evaluation, quality diagnostics, predictive maintenance indicators, and 6G communication metrics into a unified analytical dashboard. The developed Streamlit-based platform enables interactive visualization of industrial KPIs, machine-level performance analysis, efficiency distribution assessment, quality monitoring, and network diagnostics. By combining multiple operational dimensions into a single analytics environment, the proposed work provides a practical diagnostic intelligence layer that supports data-driven decision-making and operational optimization in modern smart manufacturing systems.

III. Problem Statement

Modern manufacturing systems generate large volumes of machine sensor and operational data. However, many industries still lack centralized systems capable of transforming raw industrial data into meaningful operational insights.

Manufacturing teams often face several operational challenges, including:

- Lack of centralized visibility into machine health
- Difficulty correlating sensor values with production efficiency
- Reactive handling of machine degradation and quality defects
- Inability to benchmark machine performance across operation modes
- Poor monitoring of network communication performance

As a result, operational inefficiencies may remain undetected until they significantly impact production output and manufacturing quality.

This paper aims to develop a smart factory diagnostic dashboard capable of monitoring machine health, production efficiency, quality behavior, and 6G network performance using industrial manufacturing data.

IV. Objectives

The primary objective of this paper is to develop an intelligent manufacturing analytics system capable of monitoring machine health and operational efficiency in smart factory environments.

The specific objectives are:

1) To analyze machine-level sensor data such as temperature, vibration, and power consumption.
2) To evaluate production efficiency across machines and operation modes.
3) To identify quality bottlenecks using defect rate and operational error analysis.

TABLE I: Dataset Fields Description

Column Name	Description
Machine_ID	Unique machine identifier assigned to each industrial machine.
Operation_Mode	Current operating mode of the machine such as Active, Idle, or Maintenance.
Temperature_C	Operating temperature of the machine measured in degrees Celsius.
Vibration_Hz	Machine vibration frequency measured in Hertz.
Power_Consumption_kW	Electrical power consumed by the machine in kilowatts.
Network_Latency_ms	Communication latency experienced over the 6G industrial network.
Packet_Loss_%	Percentage of communication packet loss during machine communication.
Quality_Control_Defect_Rate_%	Percentage of defective products identified during quality inspection.
Production_Speed_units_per_hr	Number of units produced per hour by the machine.
Predictive_Maintenance_Score	AI-generated score representing machine maintenance readiness.
Error_Rate_%	Percentage of operational production errors generated by the machine.
Efficiency_Status	Operational efficiency classification category: High, Medium, or Low.

4) To monitor 6G network communication parameters such as latency and packet loss.
5) To calculate industrial Key Performance Indicators (KPIs) including Machine Health Index and Error Frequency Index.
6) To develop an interactive Streamlit dashboard for real-time operational diagnostics.
7) To provide actionable recommendations for improving manufacturing efficiency and machine reliability.

V. Dataset Description

The dataset used in this paper represents industrial manufacturing process data collected from multiple smart factory machines. The dataset simulates operational records generated by Industrial IoT systems in a 6G-enabled manufacturing environment.

The dataset contains approximately 100,000 records and includes machine sensor readings, network communication parameters, production metrics, predictive maintenance indicators, quality inspection values, and operational efficiency labels. The complete dataset attributes used in this study are summarized in Table I. The dataset enables machine health diagnostics, production analysis, quality monitoring, and network performance evaluation within smart manufacturing environments.

Table I shows the industrial variables used for machine health monitoring, production analysis, quality evaluation, and operational efficiency assessment within the smart factory environment.

VI. Methodology

The paper followed a structured industrial analytics methodology consisting of multiple stages including data preprocessing, exploratory analysis, KPI generation, visualization development, and dashboard implementation.

Initially, the dataset was loaded using Python and Pandas. Data cleaning operations were performed to remove duplicate records, process date-time information, and standardize machine identifiers and operation modes.

After preprocessing, exploratory data analysis (EDA) was performed to study machine sensor behavior, production efficiency, defect patterns, and operational trends.

Several industrial KPIs were calculated to evaluate machine health and manufacturing efficiency. A Machine Health Index was created using temperature, vibration, power consumption, and predictive maintenance score.

Correlation analysis was then performed to evaluate relationships between multiple operational variables such as:

- Temperature vs defect rate
- Vibration vs operational errors
- Power consumption vs production speed
- Network latency vs efficiency status

Finally, a Streamlit dashboard was developed using Plotly visualizations to provide interactive operational diagnostics and real-time manufacturing analytics.

VII. Key Performance Indicators

Several industrial Key Performance Indicators (KPIs) were developed to evaluate machine health and operational performance.

A. Machine Health Index

The Machine Health Index represents the overall operating condition of industrial machines using temperature, vibration, power consumption, and predictive maintenance score.

B. Average Production Speed

This KPI measures the average number of units produced per hour across all machines.

C. Defect Density Score

The Defect Density Score measures product quality degradation relative to manufacturing volume.

D. Error Frequency Index

This KPI represents the operational error behavior of manufacturing machines.

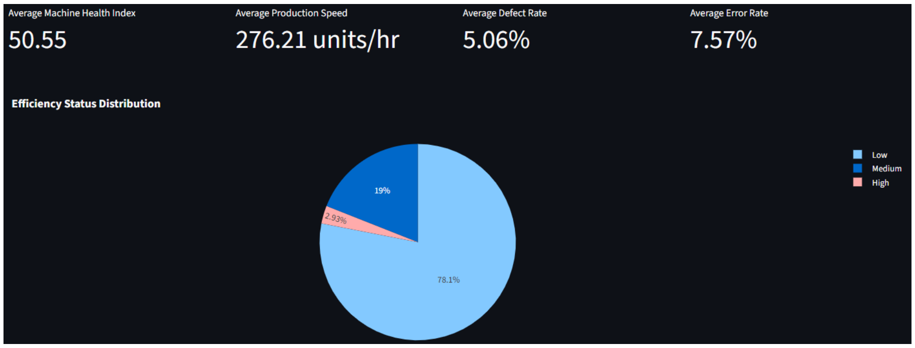


Fig. 1: Factory Health Overview Dashboard

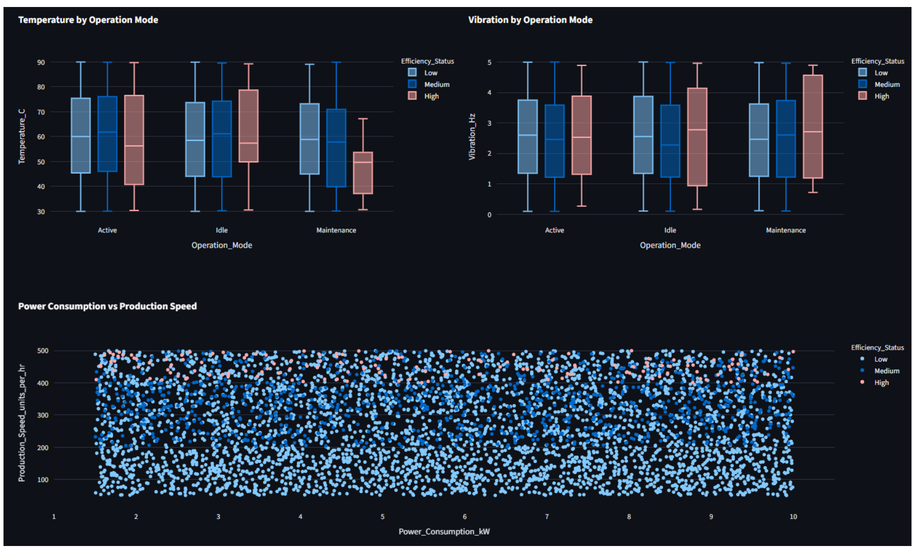
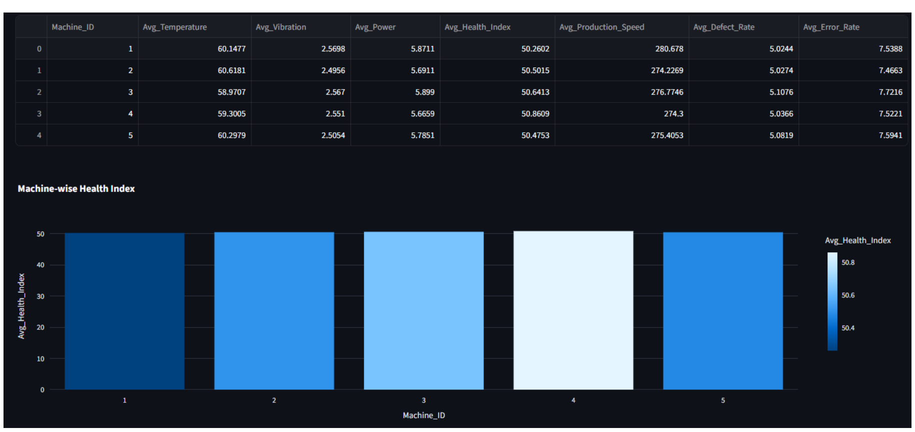


Fig. 2: Machine-Level Sensor Health Analysis



	Machine_ID	Avg_Temperature	Avg_Vibration	Avg_Power	Avg_Health_Index	Avg_Production_Speed	Avg_Defect_Rate	Avg_Error_Rate
0	1	60.1477	2.5698	5.8711	50.2602	280.678	5.0244	7.5388
1	2	60.6181	2.4956	5.6911	50.5015	274.2269	5.0274	7.4663
2	3	58.9707	2.567	5.899	50.6413	276.7746	5.1076	7.7216
3	4	59.3005	2.551	5.6659	50.8609	274.3	5.0366	7.5221
4	5	60.2979	2.5054	5.7851	50.4753	275.4053	5.0819	7.5941

Fig. 3: Machine Health Scorecard

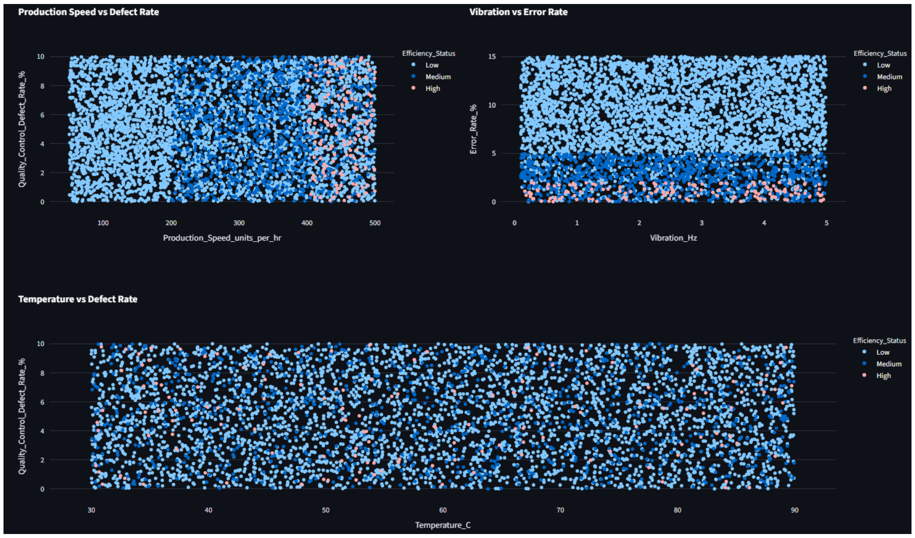


Fig. 4: Production and Quality Diagnostics

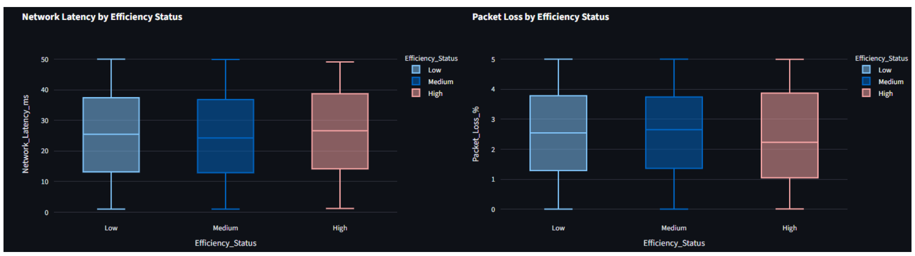


Fig. 5: 6G Network Performance Analysis

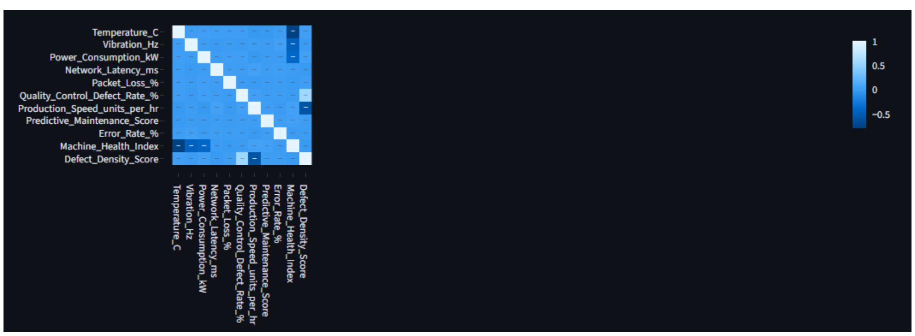


Fig. 6: Correlation Heatmap of Smart Factory Metrics

E. Efficiency Distribution

This KPI shows the percentage distribution of High, Medium, and Low efficiency operational states.

VIII. Results and Dashboard Analysis

The developed Streamlit dashboard provides centralized visibility into machine health, production efficiency, quality behavior, and network communication performance.

A. Factory Health Overview

As shown in Fig. 1, the Factory Health Overview dashboard presents the primary operational KPIs including Machine Health Index, Production Speed, Defect Rate, Error Rate, and Efficiency Status Distribution.

B. Machine-Level Sensor Health Analysis

Fig. 2 illustrates the machine-level sensor analysis dashboard, where temperature and vibration distributions are compared across different operation modes.

C. Machine Health Scorecard

Fig. 3 presents the Machine Health Scorecard, which compares machine-level health indices and operational statistics across the manufacturing environment.

D. Production and Quality Diagnostics

As shown in Fig. 4, production speed, defect rate, vibration, temperature, and operational errors are analyzed to identify quality bottlenecks and manufacturing inefficiencies.

E. 6G Network Performance Analysis

Fig. 5 presents the 6G communication performance analysis using network latency and packet loss measurements across different efficiency categories.

F. Cross-Metric Correlation Analysis

The correlation heatmap shown in Fig. 6 highlights the relationships among machine health metrics, production indicators, quality parameters, and network communication variables.

IX. System Architecture

The proposed smart factory analytics system follows a multi-stage workflow consisting of data acquisition, preprocessing, KPI generation, visualization development, and dashboard deployment. Initially, industrial machine data is collected from multiple manufacturing devices and communication systems. The collected data undergoes preprocessing and validation to remove inconsistencies and prepare the dataset for analysis. Subsequently, machine health indicators, production metrics, quality measurements, and communication parameters are analyzed to generate operational KPIs. These KPIs are then visualized using Plotly and integrated into an interactive Streamlit dashboard. The final system enables manufacturing engineers and decision-makers to monitor machine health, operational efficiency, quality performance, and network communication behavior through a centralized analytics platform.

X. Insights

The developed analytics dashboard generated several operational insights related to manufacturing efficiency and machine health behavior. The efficiency distribution analysis indicates that a large percentage of machine records fall under the Low efficiency category. This suggests that manufacturing operations may require stronger monitoring and optimization strategies. Temperature, vibration, and power consumption were identified as important indicators of machine operating condition. Machines with unstable sensor behavior may experience operational degradation and increased maintenance requirements. The quality diagnostics section demonstrates that higher vibration and temperature conditions may contribute to increased operational errors and manufacturing defects. The network analysis also demonstrates the importance of monitoring communication latency and packet loss in 6G-enabled smart manufacturing systems.

XI. Recommendations

Based on the analytical results, several recommendations are proposed:

- Monitor machines with low Machine Health Index values more frequently.
- Prioritize preventive maintenance for machines with abnormal vibration and temperature behavior.
- Investigate machines with high power consumption but lower production output.
- Improve quality monitoring systems for machines with high operational error rates.
- Continuously monitor network latency and packet loss to maintain reliable industrial communication.
- Extend the system in the future using predictive maintenance and anomaly detection algorithms.

XII. Future Work

Future enhancements may include:

- Predictive maintenance modeling using machine learning.
- Real-time streaming analytics using industrial IoT platforms.
- Anomaly detection for machine failures.
- Integration with digital twin systems.
- AI-based production efficiency forecasting.
- Advanced 6G network optimization strategies.

XIII. Conclusion

This paper successfully developed a manufacturing process health and operational efficiency analysis system for 6G-enabled smart factories. By combining machine sensor data, production metrics, predictive maintenance indicators, quality inspection values, and network communication parameters, the developed dashboard provides centralized operational visibility and supports data-driven manufacturing analytics. The Streamlit dashboard enables interactive analysis of machine health, operational efficiency, production quality, and communication performance. The paper demonstrates how industrial analytics systems can improve operational monitoring and support intelligent decision-making in modern manufacturing environments. Future work may include predictive maintenance modeling, anomaly detection systems, AI-based forecasting, and real-time industrial monitoring solutions.

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Dakshanya Maddala received the Bachelor’s degree in Electronics and Communication Engineering from Jawaharlal Nehru Technological University, Kakinada, India, in 2020, and the Master’s degree in Computer Science and Engineering from Texas A&M University–Kingsville (TAMUK), USA, in 2023. She is currently pursuing the Ph.D. degree in Electrical Engineering and Computer Science at Texas A&M University–Kingsville, where she serves as a Graduate Research Assistant. Her research interests include smart manufacturing systems, Industrial Internet of Things (IIoT), wireless sensor networks, machine learning, computer vision, autonomous systems, robotics, intelligent monitoring systems, and secure IoT architectures. Her current research focuses on applying data-driven methodologies and artificial intelligence techniques to improve operational efficiency, machine intelligence, autonomous decision-making, and industrial analytics in next-generation smart environments. She has authored and co-authored multiple research publications in international conferences and journals in the areas of wireless sensor networks, intelligent coverage systems, IoT-enabled architectures, and AI-driven analytics. She has also contributed as a reviewer for IEEE conferences and technical symposia. In addition to her academic research, she has gained industry experience through internships and applied research projects involving machine learning, data analytics, predictive modeling, industrial monitoring systems, and intelligent automation. Her long-term research vision is to bridge artificial intelligence, industrial automation, robotics, and next-generation communication systems to develop intelligent and sustainable cyber-physical systems for future smart industries.